Day 03

P-values, P-hacking, and Multiple testing

- What P-values measure (and what they don't)
- P-hacking: subtle and obvious forms
- Multiple testing problem
- Publication bias and file drawer problem
- Conflicts of interest and analytical flexibility
- Common interpretation errors

Statistical hypothesis testing

Abstract

Formula display: MathJax

Background

Many groups, including our own, have proposed the use of DNA methylation profiles as biomarkers for various disease states. While much research has been done identifying DNA methylation signatures in cancer vs. normal etc., we still lack sufficient knowledge of the role that differential methylation plays during normal cellular differentiation and tissue specification. We also need thorough, genome level studies to determine the meaning of methylation of individual CpG dinucleotides in terms of gene expression.

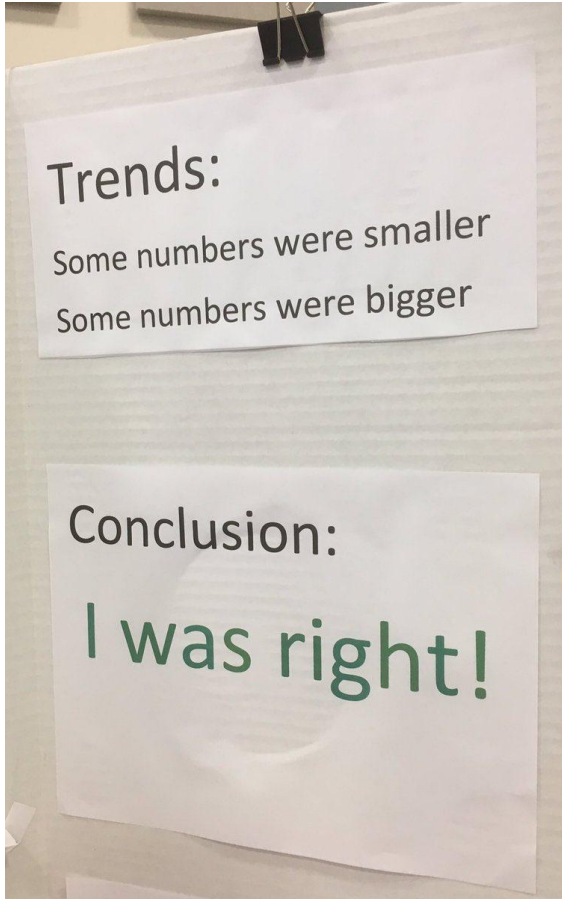
Results

In this study, we have used (insert statistical method here) to compile unique DNA methylation signatures from normal human heart, lung, and kidney using the Illumina Infinium 27 K methylation arraysand compared those to gene expression by RNA sequencing. We have identified unique signatures of global DNA methylation for human heart, kidney and liver, and showed that DNA methylation data can be used to correctly classify various tissues. It indicates that DNA methylation reflects tissue specificity and may play an important role in tissue differentiation. The integrative analysis of methylation and RNA-Seq data showed that gene methylation and its transcriptional levels were comprehensively correlated. The location of methylation markers in terms of distance to transcription start site and CpG island showed no effects on the regulation of gene expression by DNA methylation in normal tissues.

Conclusions

This study showed that an integrative analysis of methylation array and RNA-Seq data can be utilized to discover the global regulation of gene expression by DNA methylation and suggests that DNA methylation plays an important role in normal tissue differentiation via modulation of gene expression.

https://nsaunders.files.wordpress.com/2012/07/bmcsysbiol.png



Statistical hypothesis testing

Many scientific studies are interested in quantifying the difference in a particular parameter b/w two groups.

- There's always some difference → Is it statistically significant difference?

Say you're testing the efficacy of a cold medicine:

- Two groups: one given a placebo & the other given a drug
- Followed-up: how long the cold lasted in each person in both groups
- Null: Ineffective; Alternative: Effective

Statistical hypothesis testing

1. **Decide on the effect** that you are interested in, design a suitable experiment or study, pick a data summary function and test statistic.
2. **Set up a null hypothesis** , which is a simple, computationally tractable model of reality that lets you compute the null hypothesis.
3. **Decide on the rejection region** , i.e., a subset of possible outcomes whose total probability is small.
4. **Do the experiment** and collect the data, compute the test statistic.
5. **Make a decision** : reject the null hypothesis – i.e. conclude that it is unlikely to be true – if the test statistic is in the rejection region.

Statistical hypothesis testing

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Null hypothesis	**Null distribution**
Typically a statement of no relationship between variables or no effect of an experimental manipulation/intervention.	The set of possible outcomes of the test statistic (& their probabilities) under the assumption that the null hypothesis is true.

Statistical hypothesis testing

S: I just love making people laugh. I think humor is like medicine!
N: Oh, we must be in the placebo group.

- **Two groups:** Drug vs Placebo
- **Measurement:** *Difference* in #healthy-days
- **Null:** Ineffective | **Alternative:** Effective

Statistical hypothesis testing

Raw Data		Mean
Drug (group1)	0, 5, 2, −2, 8, −6, 0, 0, −6, −3, −7, 4, −2, −2, 0, −4, −1, −8, 0, 6	−0.80
Placebo (group2)	−2, 2, −2, −5, 7, 11, 6, 2, 1, 1, 6, −2, −1, 7, 7, −2, 4, 3, −1, 17	2.95

Quantifying the difference between the groups	
Effect size | Diff. b/w the means of two groups $(\mu_2 - \mu_1)$	3.75
Test statistic | $(\mu_2 - \mu_1) / \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$	2.45

What is the P-value?

Study: If **a specific drug** is **effective** in reducing the symptoms of a **disease**.

Given this context, which of the following defines **what is P-value**?

1. The **amount of evidence** that the drug is effective against the disease.
2. The **strength of the effect** of the drug against the disease.
3. One minus the probability that the **outcome** of the study **is important**.
4. One minus the probability that the result can be **replicated**.
5. One minus the probability that the **drug is effective** against the disease.
6. The probability that the **drug is not effective** against the disease.

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- We observe a difference between the groups.
- But, how *unlikely* is the observed difference

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Null hypothesis:
The drug is ineffective.

- Even if the drug is only as good as the placebo, we'll observe as much or more difference between the two groups.
- Irrespective of the group each individual is a part of, we'll observe as much or more difference between the two groups.

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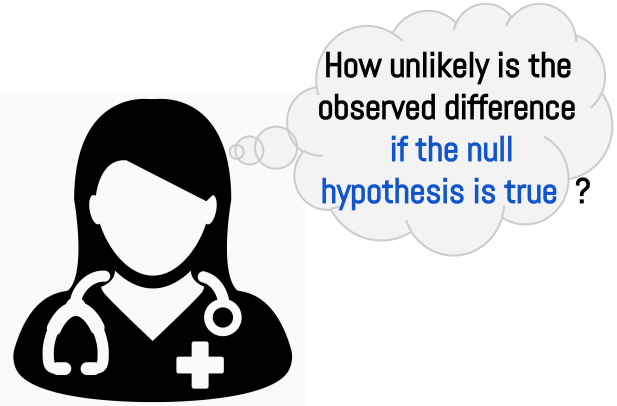

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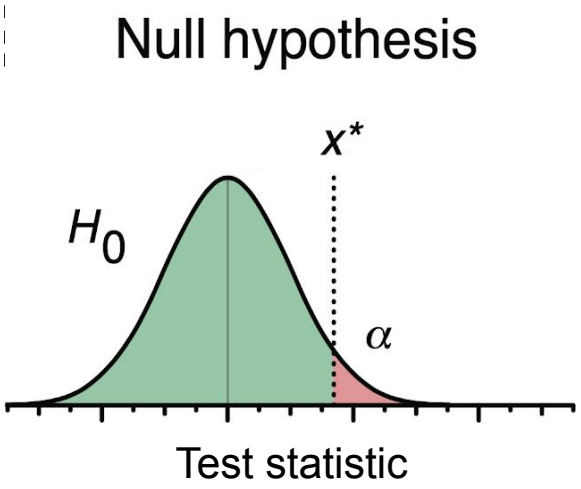
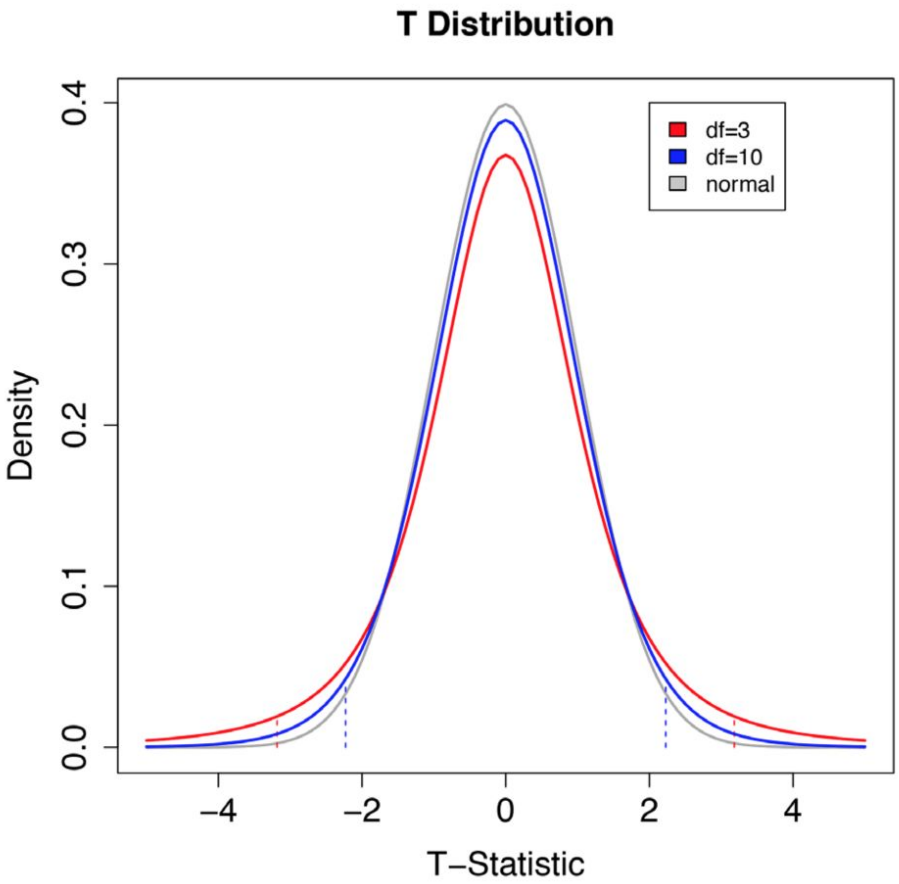
How unlikely is the observed difference if the drug ≈ placebo?

1. Calculate the test statistic from the real data.
2. Repeat the following 10,000 times to set up the null hypothesis for this test statistic:
   - Randomly assign individuals to groups.
   - Record the test statistic from the permuted data.
3. Calculate how unlikely the real test statistic is.



The **p-value** is the probability that the experiment would have produced the observed outcome — or something more extreme — if the drug were ineffective.

P-value



The p-value is the area under the null distribution corresponding to outcome equal to or more extreme than the observed statistic.

Student's one-sample test

$$t = \frac{\bar{x} - \mu_0}{\text{SEM}}$$

Welch's two-sample test

$$t = \frac{\overline{X}_1 - \overline{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}$$

P-value

History

Fisher (1920s):

- Informal method to help interpret the data along with prior experience, domain knowledge, size of the effect, etc.

Neyman & Pearson:

- Control false positive rate at α, set by the experimenter based on what can be tolerated.
- Formulate null and alternative hypothesis.
- Reject null when $p < \alpha$.
    - The threshold $\alpha = 0.05$ is merely a convention.

P-value

Type I & type II errors

P-value captures if there is “sufficient” inconsistency with the null hypothesis.

Choosing $p < \alpha$ controls type I error at α.

- Type I error: False-positive rate (α)
- Type II error: False-negative rate (β)
- Remember the story of the boy that cried wolf!

Remember, mixing up Type I and Type II errors is called a Type III error

Giving mistakes numbers instead of names was a real Type IV error

P-value

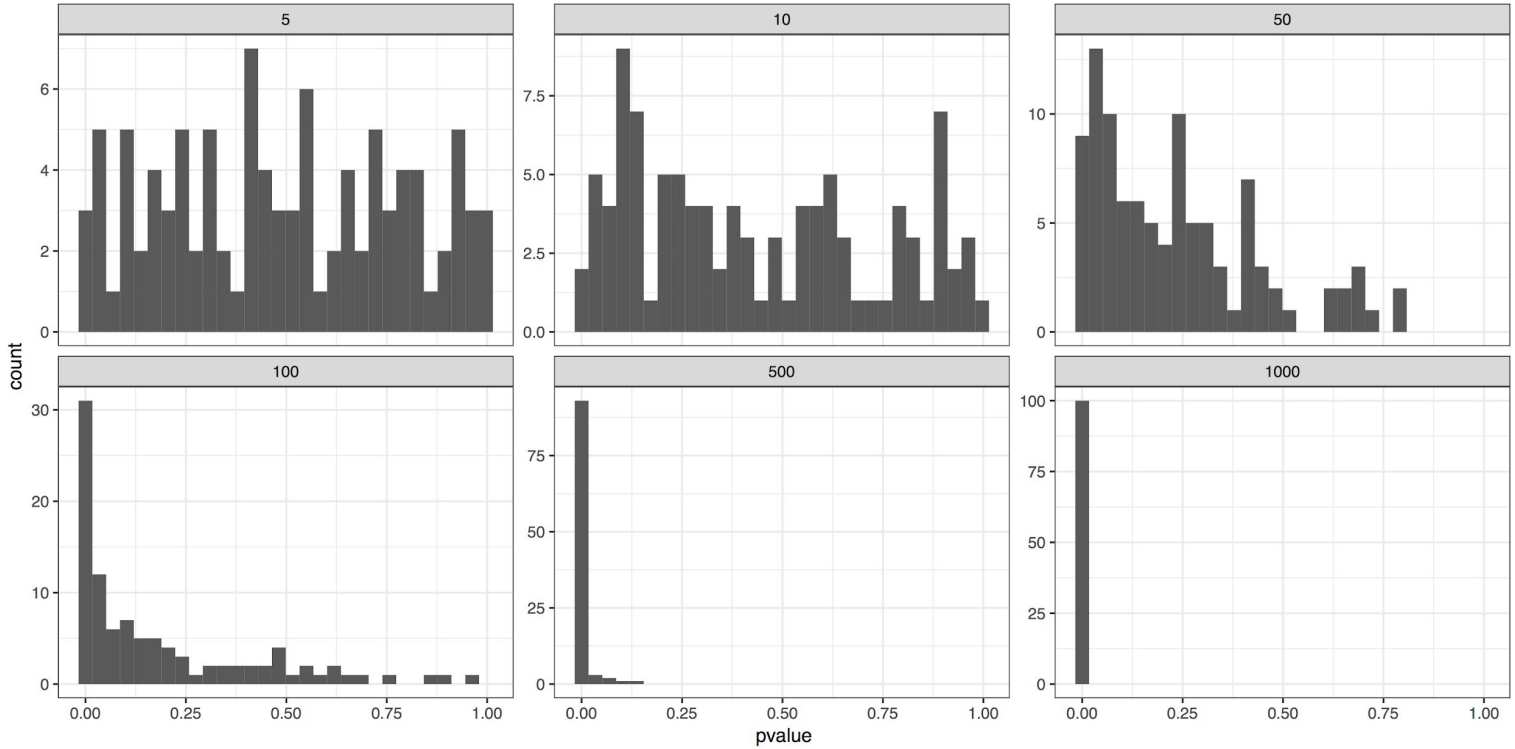
P-values are dependent on :

- Size of the effect (effect size)
- Variance within each group
- Sample size
- The underlying experimental design & the null hypothesis (need not always be random chance).

  a. Conversely, two completely different experiments can give same data but end up very different p-values.
     - 3 out of 9: Binomial p-value = 0.073
     - 3 out of 9: Neg. Binomial p-value = 0.033.

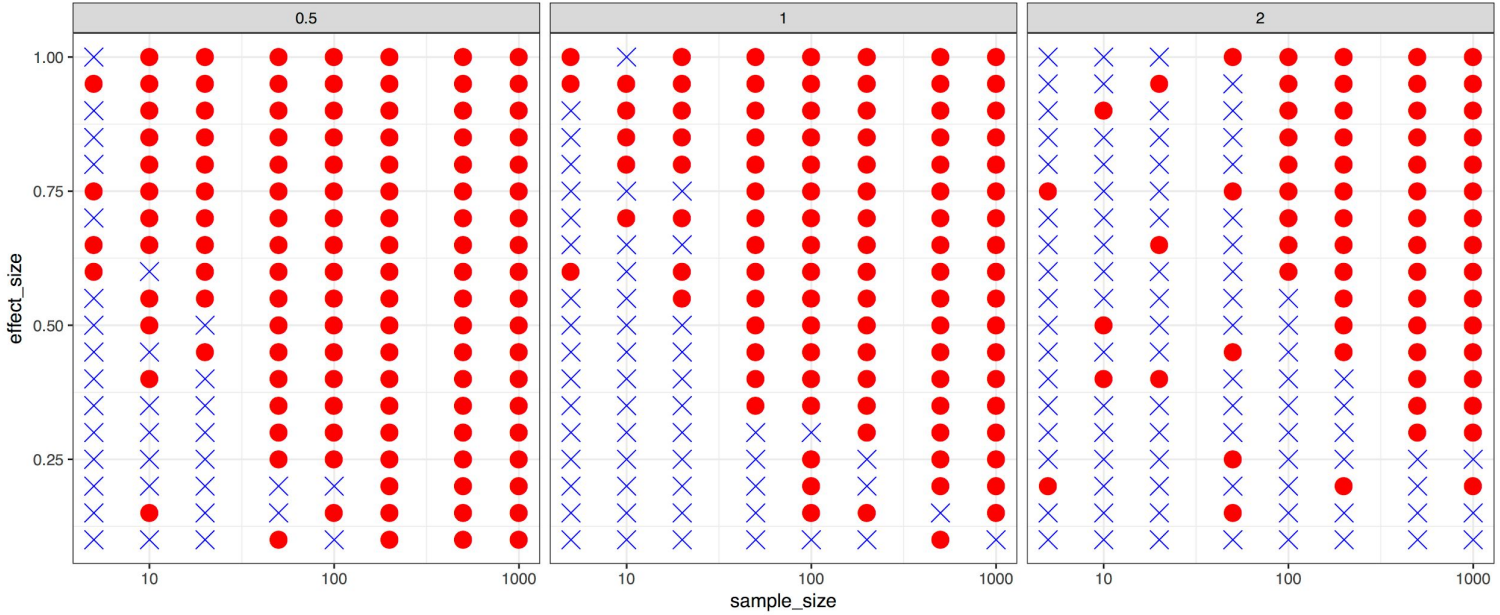
P-value

P-values are dependent on : sample_size (effect_size = 0.25, std_deviation = 1)



P-value

P-values are dependent on : sample_size, effect_size, within-group variance



Just collecting more data can help reach smaller P-values.

P-value

Significant or not!

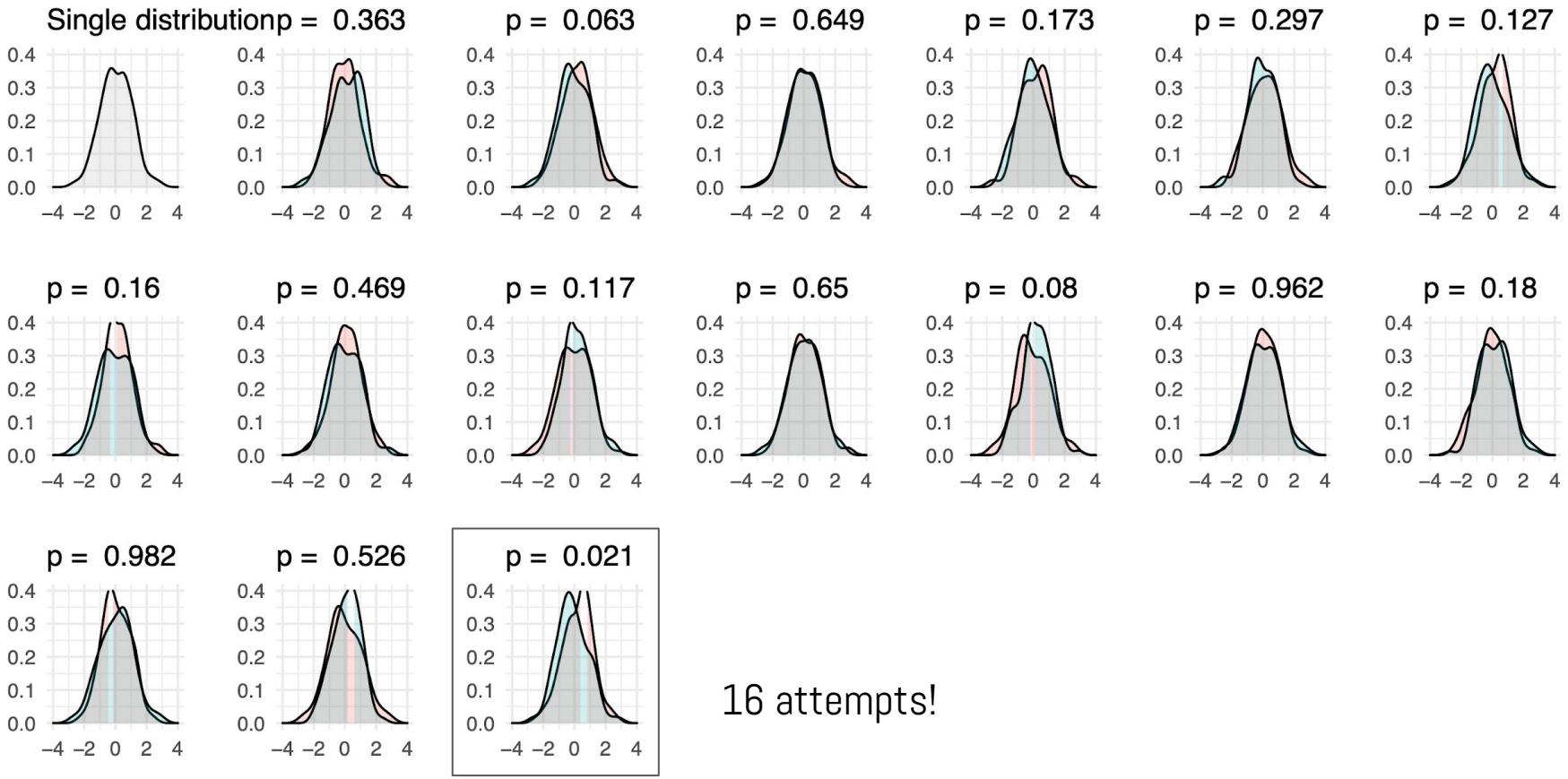
https://mchankins.wordpress.com/2013/04/21/still-not-significant-2/

The following list is culled from peer-reviewed journal articles in which:

A. The authors set themselves the threshold of 0.05 for significance,
B. Failed to achieve that threshold value for p, and
C. Described it in such a way as to make it seem more interesting.

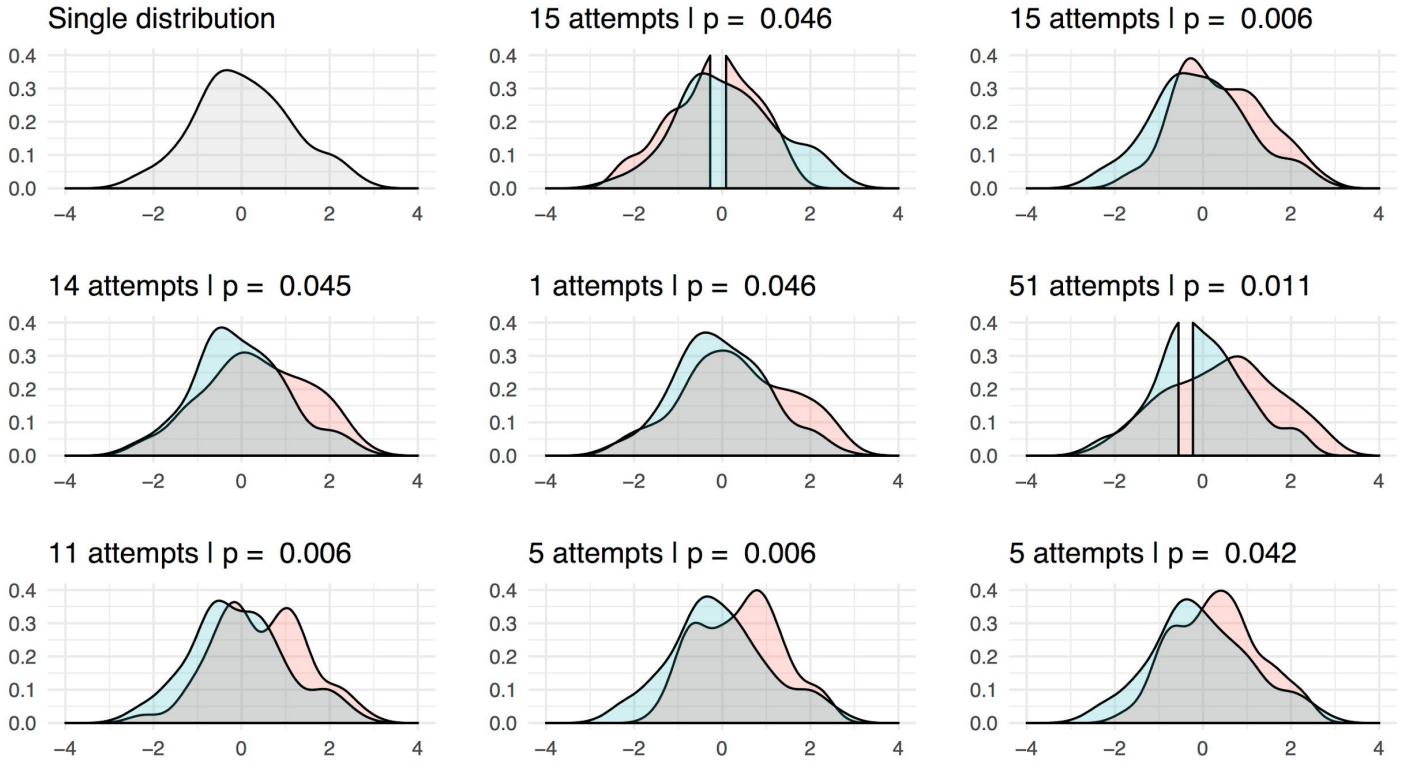
(barely) not statistically significant (p=0.052)
a barely detectable statistically significant difference (p=0.073)
a borderline significant trend (p=0.09)
a certain trend toward significance (p=0.08)
a clear tendency to significance (p=0.052)
a clear trend (p<0.09)
a clear, strong trend (p=0.09)
a considerable trend toward significance (p=0.069)
a decreasing trend (p=0.09)
a definite trend (p=0.08)
a distinct trend toward significance (p=0.07)
a favorable trend (p=0.09)

P-value



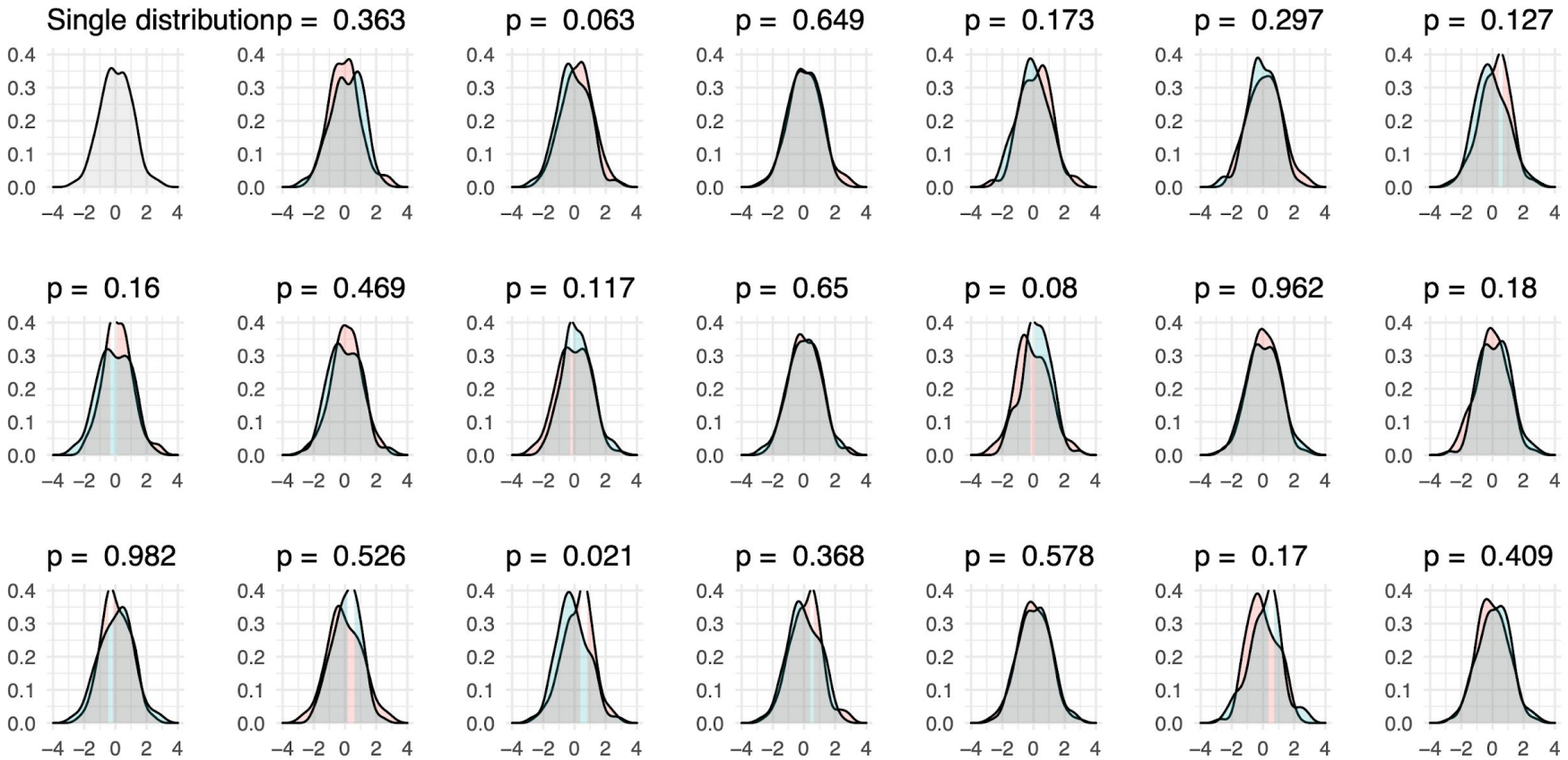
16 attempts!

P-value



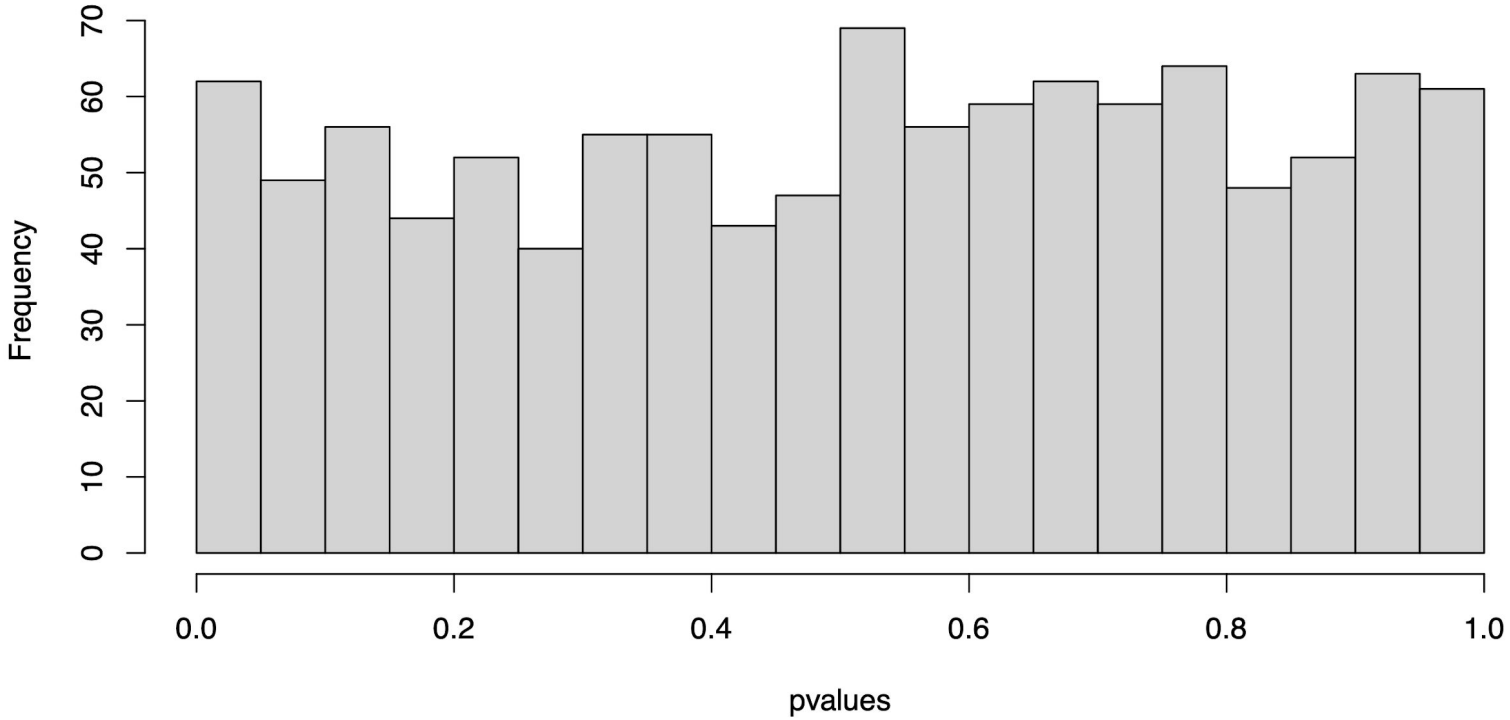
"When a measure become a target, it ceases to be a good measure" – Srathern's Law

P-value


Single distribution
p = 0.363
p = 0.063
p = 0.649
p = 0.173
p = 0.297
p = 0.127
p = 0.16
p = 0.469
p = 0.117
p = 0.65
p = 0.08
p = 0.962
p = 0.18
p = 0.982
p = 0.526
p = 0.021
p = 0.368
p = 0.578
p = 0.17
p = 0.409
0.4
0.3
0.2
0.1
0.0
−4 −2 0 2 4

P-value

Distribution of p-values under the null hypothesis



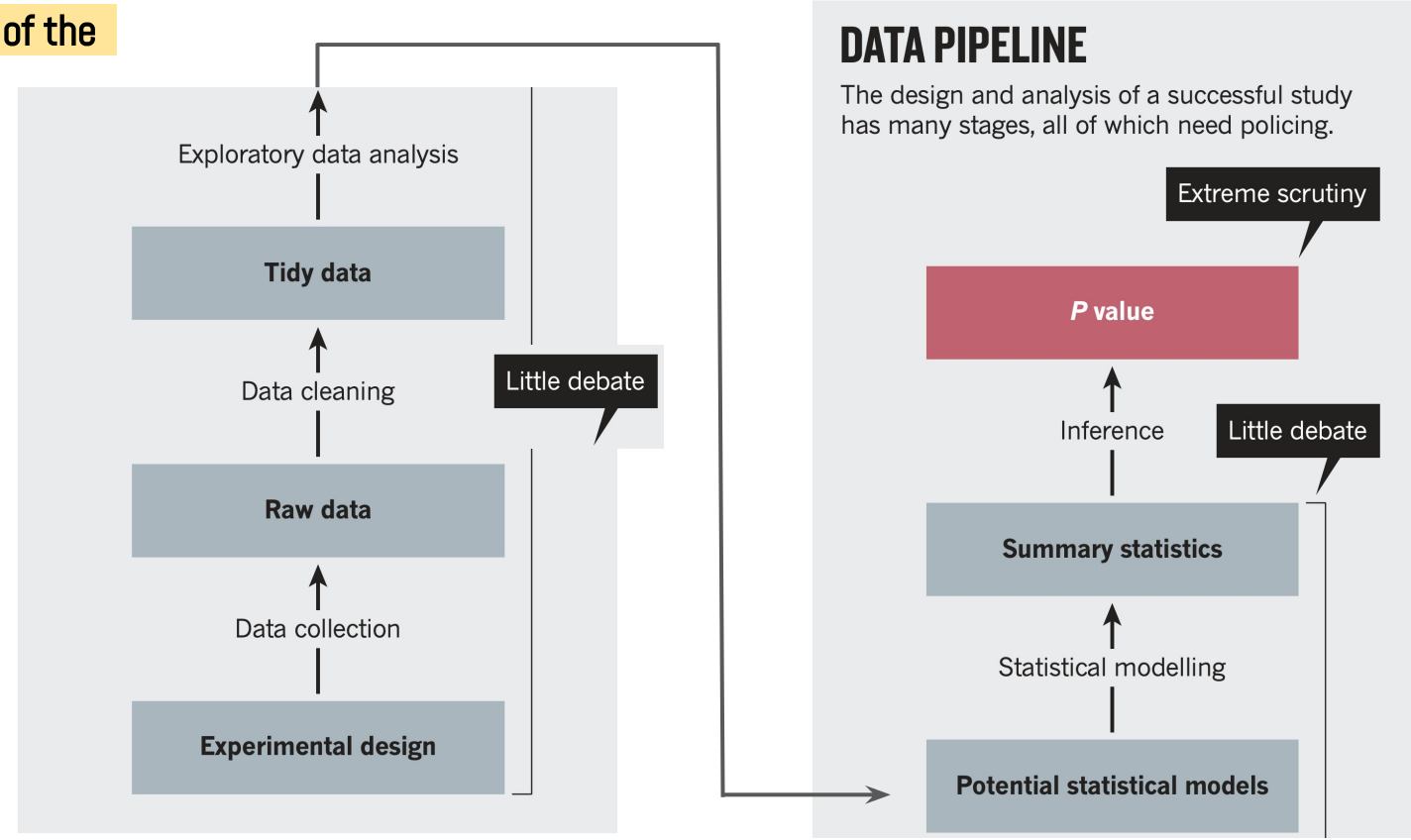
P-value

Common misinterpretations

- Inverse probability error: A p-value is not the probability that the null hypothesis is true, nor is 1−p the probability that the alternative hypothesis is true.
- P-value ≠ Effect size: P-values do not reflect the strength or magnitude of a relationship.
- Statistical significance ≠ Practical importance: This is often called the “sizeless stare”.
- P-value > 0.05 ≠ "No effect": A non-significant result is not proof that the null hypothesis is true; it merely means the evidence was insufficient to reject it.

P-value

Just the tip of the iceberg!



JT Leek, RD Peng http://www.nature.com/news/statistics-p-values-are-just-the-tip-of-the-iceberg-1.17412

P-hacking

Researcher degrees of freedom: Also known as the "garden of forking paths," the near-endless number of analytical choices a researcher can make (e.g., excluding certain outliers, choosing specific transformations) that can all lead to different results from the same dataset.

P-hacking: Collect or select data or statistical analyses until nonsignificant results become significant.

- Conduct analyses midway through experiments to decide whether to continue collecting data.
  - Collect data until the p-value drops below 0.05 and then stop.
  - Stop data exploration if an analysis yields a significant p-value.
- Record & test many response variables and decide which to report post-analysis
- Decide whether to include or drop outliers post-analyses (after seeing how they affect p-values)
- Exclude, combine, or split treatment groups post-analysis (i.e., where an effect appears significant)
- Include or exclude covariates post-analysis
- Repeatedly try log-transformations, square roots, or other methods until significance is manufactured

P-hacking

Researcher degrees of freedom: Also known as the "garden of forking paths," the near-endless number of analytical choices a researcher can make (e.g., excluding certain outliers, choosing specific transformations) that can all lead to different results from the same dataset.

The trap of validity

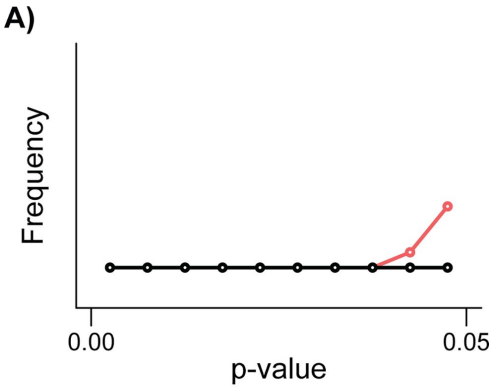
These practices often feel like “valid exploration” or “improving the model” to the researcher, which makes them particularly dangerous and common.

Detection

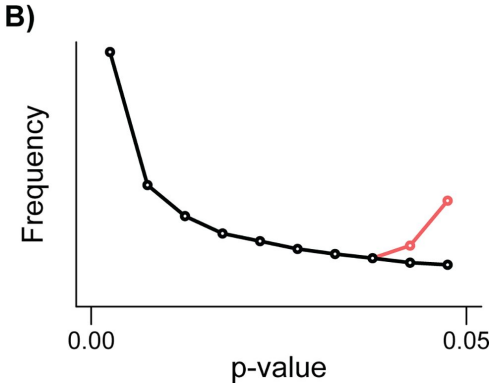
Look for impossibly high effect sizes or “new” variables appearing in the results that weren’t clearly defined in the methodology.

P-hacking

No evidence of effect

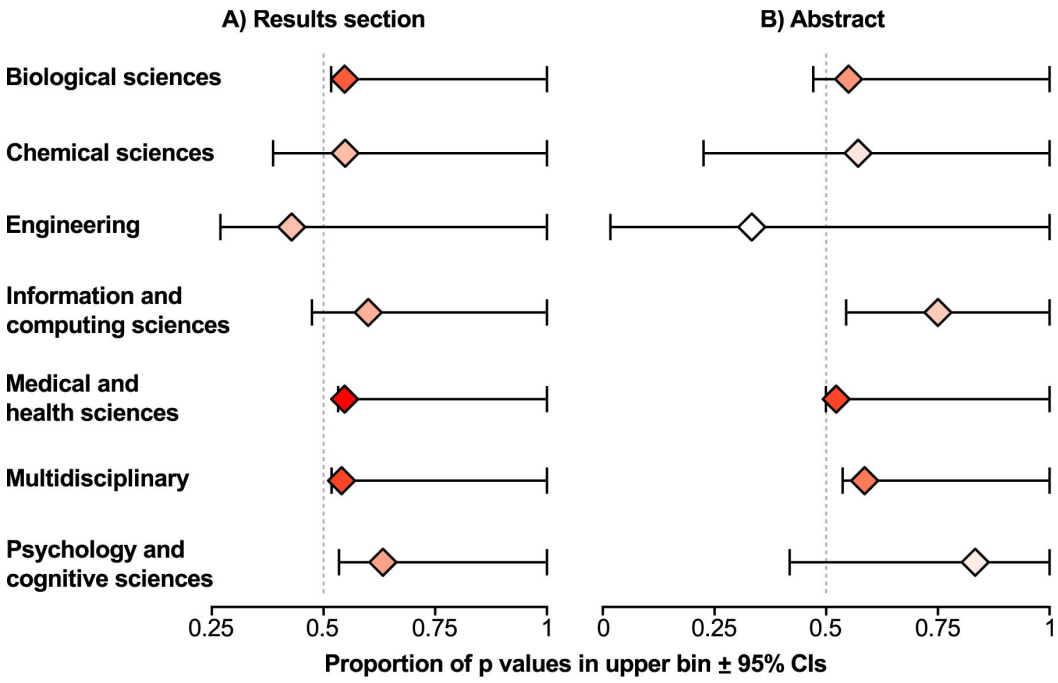


Evidence of effect



The Extent and Consequences of P-Hacking in Science https://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.1002106

P-hacking



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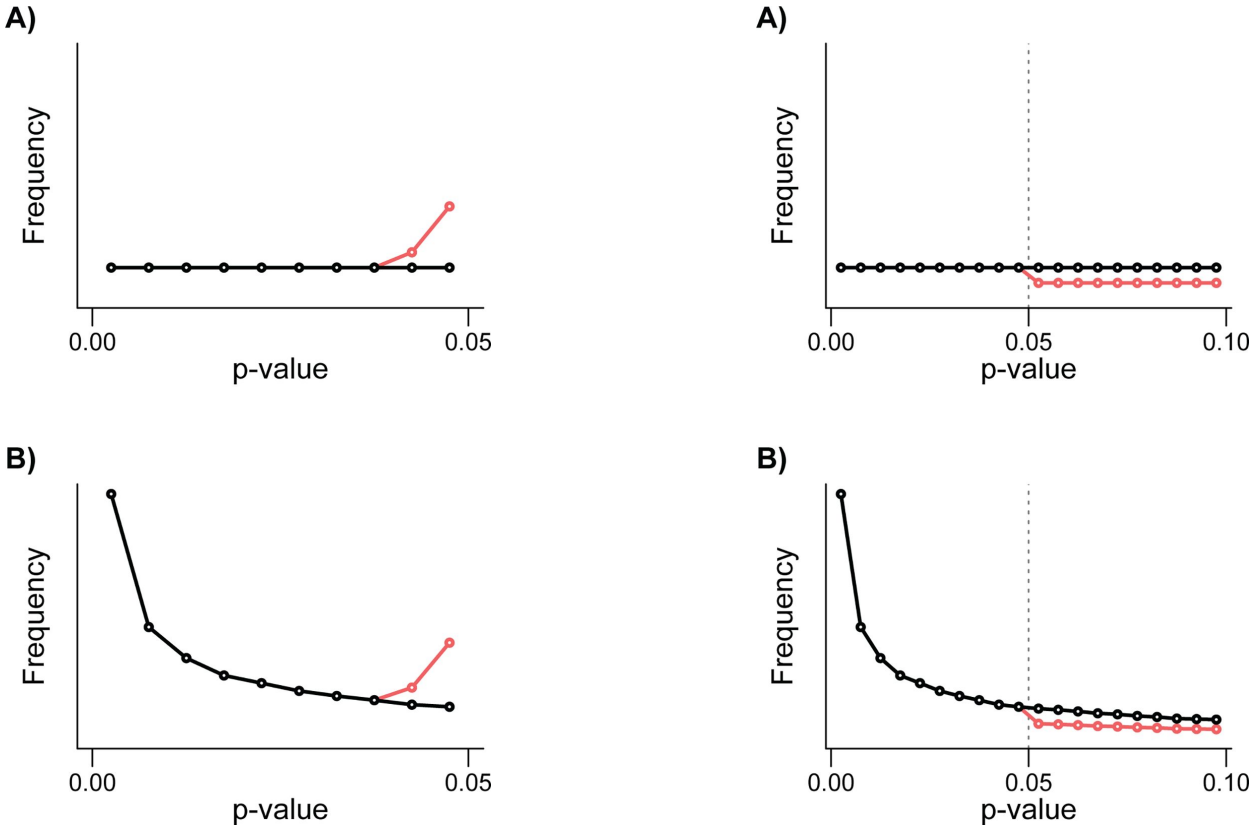
Publication bias

No evidence of effect

Evidence of effect



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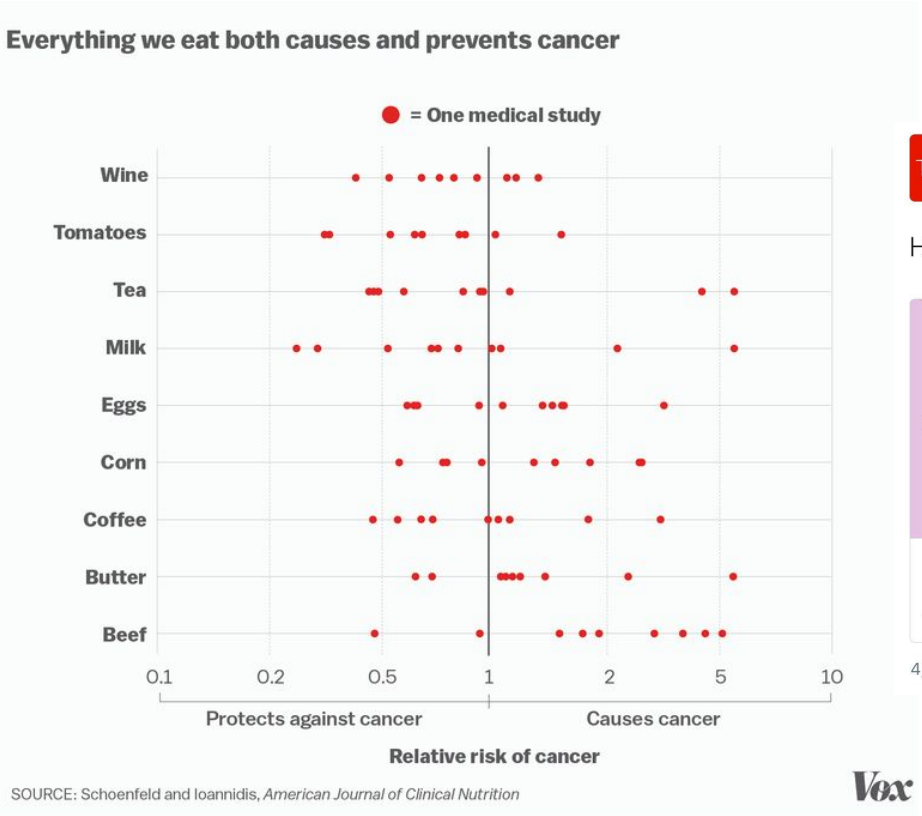
Publication bias

Literature distortion

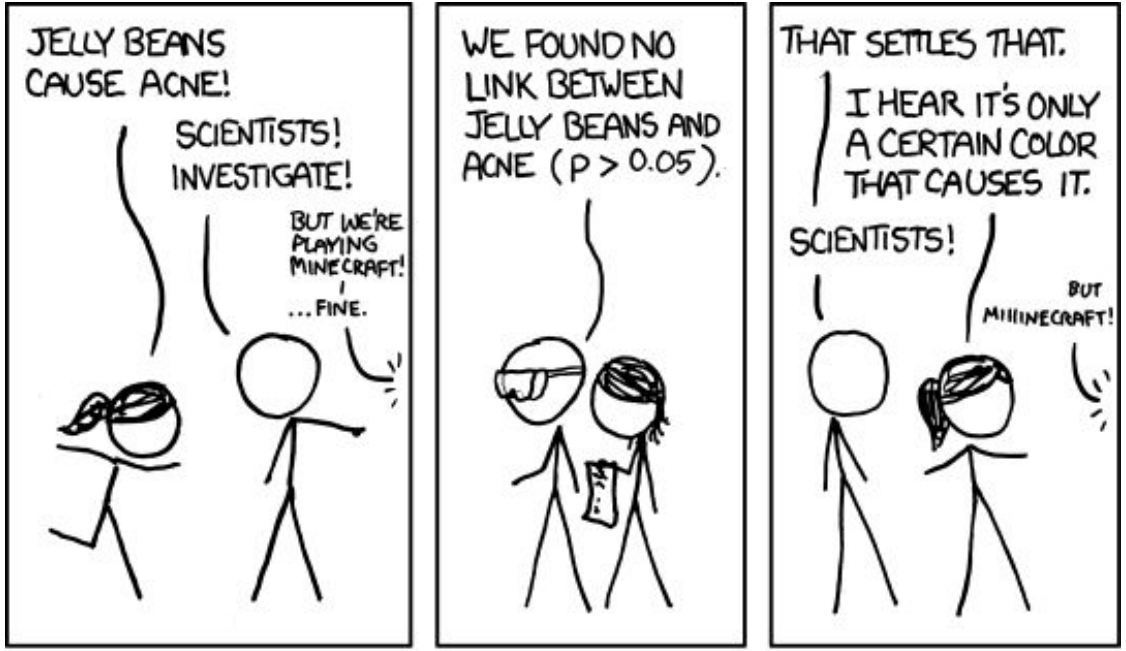
Journals and reviewers strongly prefer “significant” novelties over “boring” inconclusive results, causing the literature to be filled with the 5% of studies that show Type I errors while the other 95% stay in researchers’ file drawers.

Winner’s curse: Because lucky fluctuations into significance (in underpowered studies) are published, effect sizes in the literature are systematically overestimated.

Multiple hypothesis testing



Multiple hypothesis testing



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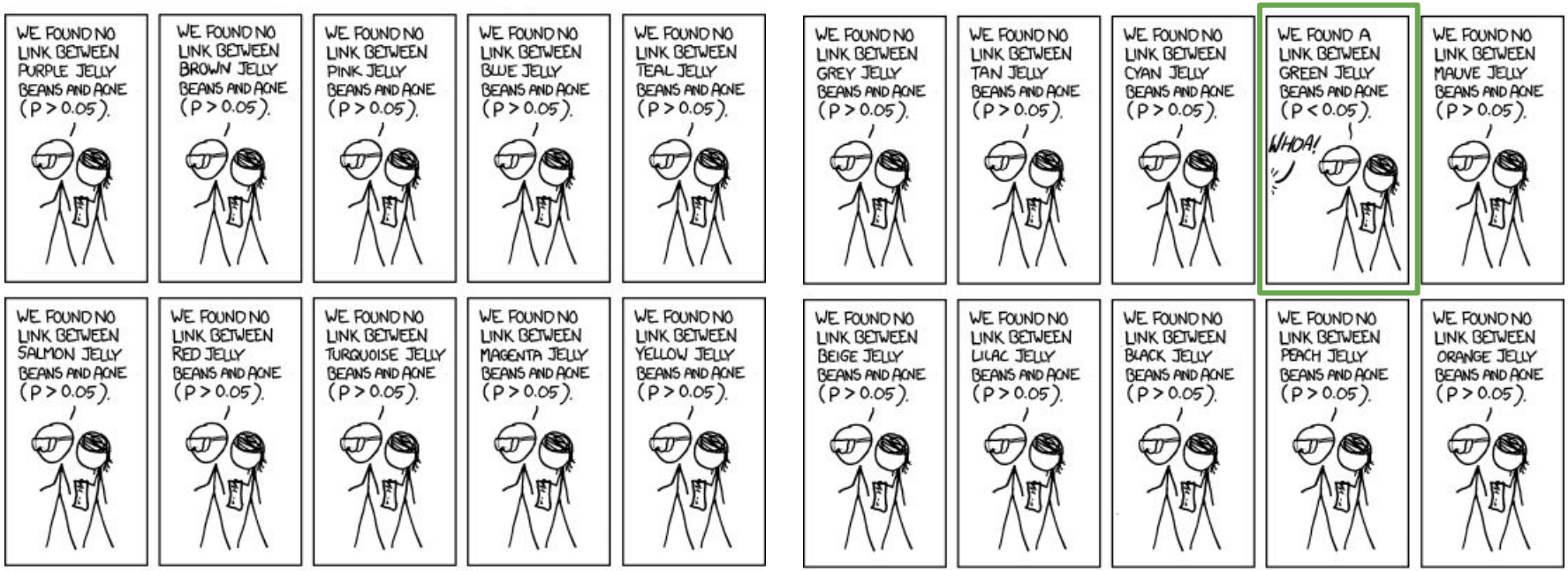

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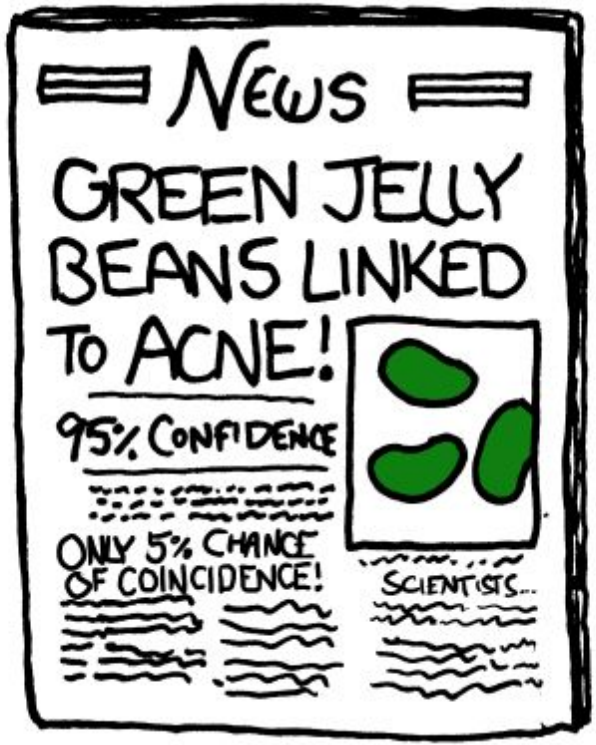

The more inferences are made, the more likely erroneous inferences are to occur.

- Testing 10 independent null hypotheses on one dataset creates a 40% chance of finding at least one “significant” result by pure luck.

Not all variables require correction; e.g., control variables included to adjust for confounding do not typically count as independent hypothesis tests.

Multiple hypothesis testing

Correction methods

These techniques generally require a **stricter significance threshold for individual comparisons**, so as to compensate for the number of inferences being made.

Family-wise error rate (FWER)

Aims to prevent even a single false rejection within a "family" of related tests.

Bonferroni correction

Dividing the α (usually 0.05) by the number of tests. Simple but highly conservative, often failing to detect real effects in large variable sets.

False discovery rate (FDR)

(E.g. Benjamini-Hochberg) Offer more power by controlling the proportion of false positives among all rejected null hypotheses.

Which one to choose? Rigor, study type